



MUSCULOSKELETAL INFECTIONS- PART A BONE AND JOINT INFECTIONS

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LEARNING AND TESTABLE OBJECTIVES

- Part A-List the major pathogens of osteomyelitis, arthritis and myositis.
- Describe the routes of infection
- For the specific etiologic agents of musculoskeletal infections, discuss pertinent characteristics:
 - Morphological and cultural characteristics
 - Reservoir and/or source
 - Transmission and relevant epidemiology
 - Pathogenesis and virulence factors
 - Disease association and complications
 - Laboratory Diagnosis
 - Treatment, Prevention

OSTEOMYELITIS AND INFECTIOUS ARTHRITIS



OSTEOMYELITIS AND ARTHRITIS

Examples of commonly associated microbes

Osteomyelitis : An infection of the bone involving the periosteum, cortical bone and the medullary cavity.



Most common cause is ***Staphylococcus aureus***

Pyogenic/Septic arthritis is a purulent infection of the joint- usually bacterial.



Eg. Arthritis due to **staphylococcal or gonococcal infection**

Reactive arthritis: Post infectious arthritis without direct bacterial invasion- immune mediated



Eg. Arthritis due to **genital Chlamydial infection**

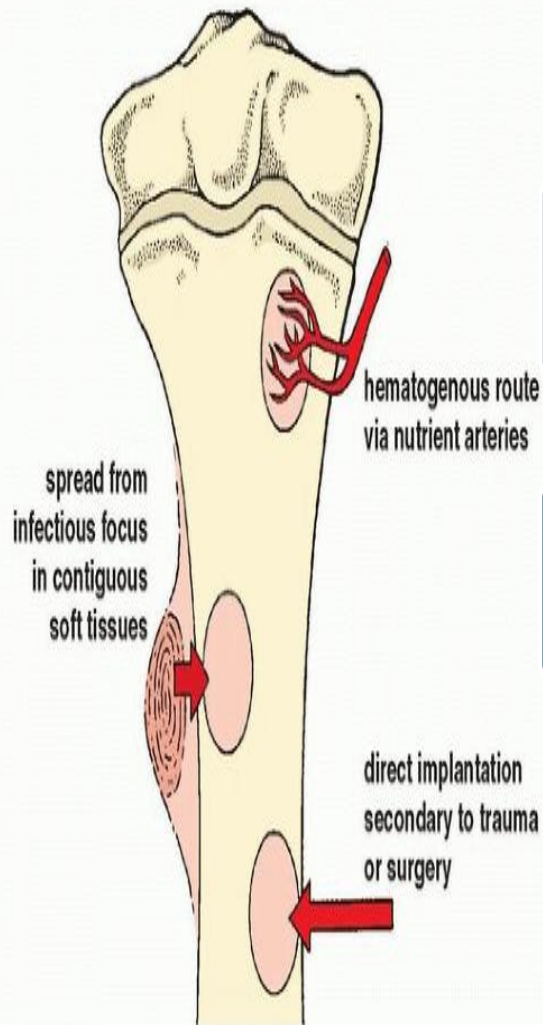
Viral arthritis: Usually seen in the prodromal period of some viral infections. Due to inflammation by the virus or due to deposition of circulating immune complexes(IC).



Eg. Immune complex mediated arthritis seen as extra hepatic manifestations in **Hepatitis B / Hepatitis C infections**

OSTEOMYELITIS/INFECTIOUS ARTHRITIS

ROUTES OF INFECTIONS



Hematogenous spread

Usually monomicrobial

Contiguous spread

Associated with direct inoculation

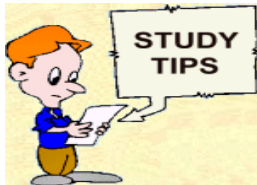
Trauma/ Surgery

Associated with vascular insufficiencies

Diabetics

Often polymicrobial

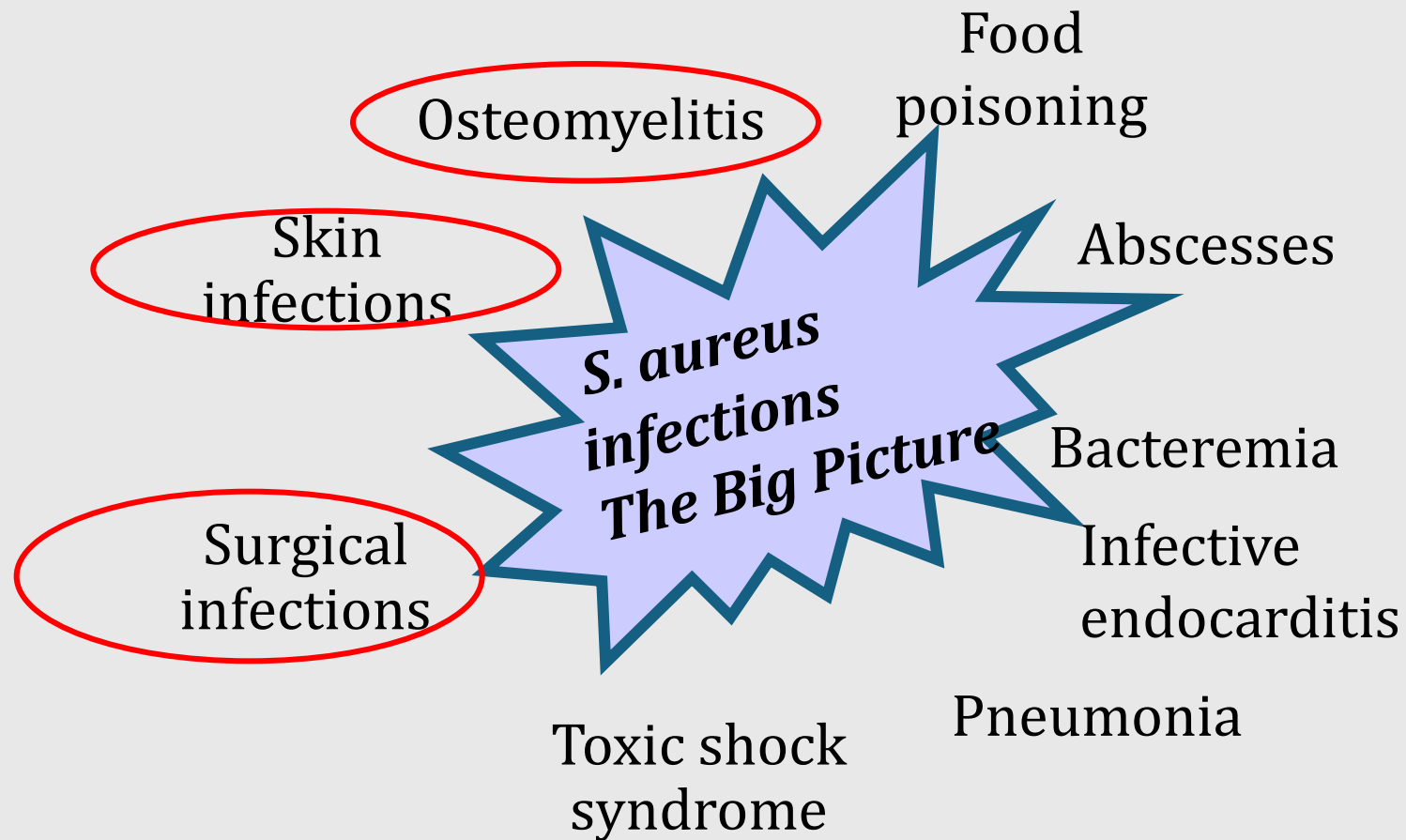
MICROBES CAUSING OSTEOMYELITIS



ETIOLOGY OF OSTEOMYELITIS/ ARTHRITIS



Setting/predisposing factor	Etiology
No other information available/ Most common	<i>Staphylococcus aureus</i> (#1 cause of osteomyelitis)
Prosthetic joint replacement	<i>S. aureus, S. epidermidis</i> (biofilms)
IV drug users	<i>Pseudomonas, Candida, S. aureus</i>
Diabetic foot	<i>Pseudomonas, E.coli, Serratia etc.</i>
Sickle cell	<i>Salmonella</i>
Vertebral involvement	<i>Mycobacterium tuberculosis</i> (Pott's spine)
Tick Bite	<i>Borrelia burgdorferi</i>
Sexually active with arthritis	<i>Neisseria gonorrhoeae</i> <i>Chlamydia trachomatis</i>
Post-diarrheal Reactive arthritis	<i>Shigella, Salmonella, Campylobacter,</i> <i>Yersinia</i>



STAPHYLOCOCCUS AUREUS:

MOST COMMON CAUSE OF OSTEOMYELITIS AND INFECTIOUS ARTHRITIS

Present on **human skin and nasal mucosa**
Spread from **person-to-person** by direct contact

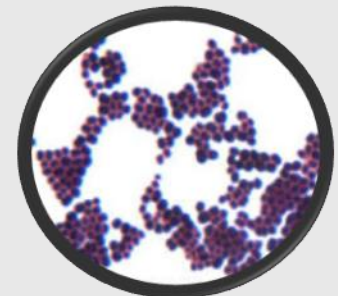


- **GPC**
- **Arranged in “grape-like” clusters**
- **Catalase (+)**
- Salt tolerant
- Facultative anaerobe

All staphylococci

- **Coagulase (+)**
- **Mannitol (+)**
- β -hemolysis
- Golden colonies

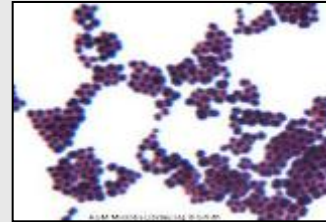
Specific to *S. aureus*



S. aureus MORPHOLOGY AND IDENTIFICATION

■ Gram stain

- Gram-positive cocci in clusters



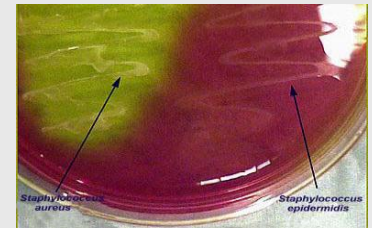
■ Culture

- β -hemolytic on blood agar
 - Golden pigmented colonies

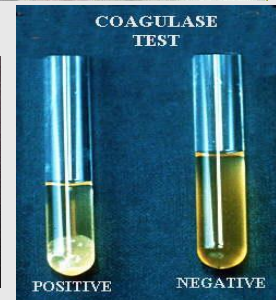


- Mannitol salt agar

- 7.5% NaCl ➔ selects for Staphylococci
- fermentation of mannitol differentiates *S. aureus* from other staphylococci

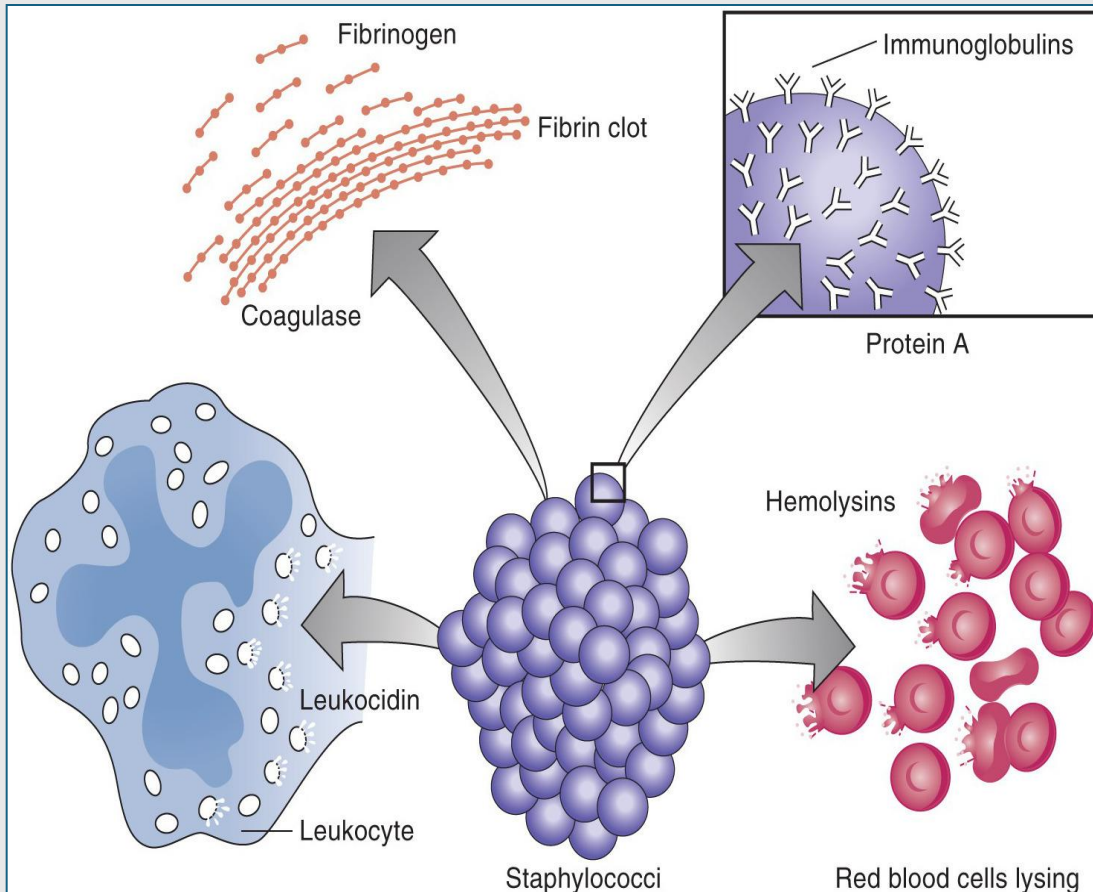


- Catalase (substrate- H_2O_2) and coagulase (plasma) tests positive



Staphylococcus aureus

VIRULENCE FACTORS



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○ **Surface Adhesins**

- **Protein A** – binds Fc portion of antibody, inhibits phagocytosis
- **Collagen binding and fibrinogen binding proteins**

○ **Coagulase** – converts fibrinogen to a fibrin clot

○ **Cytolytic toxins – Hemolysins and Panton Valentine Leukocidins** – pore-forming toxins

○ **Hyaluronidase** – breaks down connective tissue

VIRULENCE FACTORS

Surface adhesins and Biofilm formation

Surface adhesins(Collagen binding and Fibronectin binding proteins) bind microbes to bone matrix.

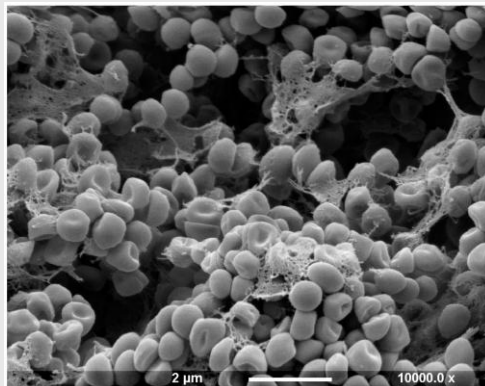
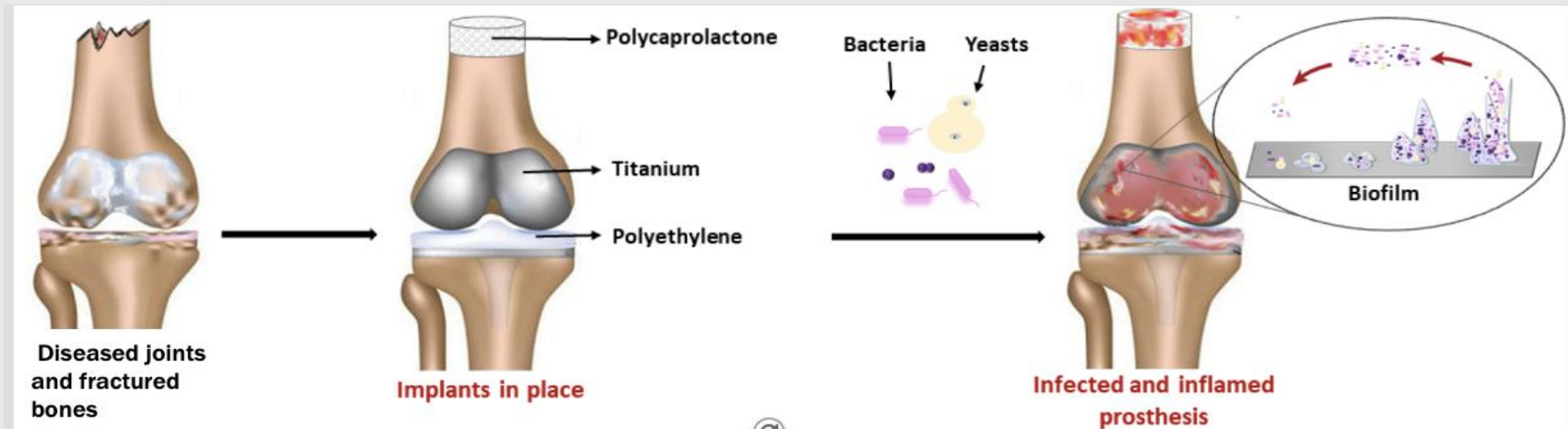


Figure-Scanning electron micrograph of a six-day-old Staphylococcus biofilm demonstrating bacteria within an extracellular polymeric substance.

Ref: Rodríguez-Merchán, E.C.; Davidson, D.J.; Liddle, A.D. Recent Strategies to Combat Infections from Biofilm-Forming Bacteria on Orthopaedic Implants. *Int. J. Mol. Sci.* **2021**, *22*, 10243. <https://doi.org/10.3390/ijms221910243>

S. aureus INFECTIONS - TREATMENT

- Treatment guided by susceptibility testing
 - Methicillin-sensitive strains (MSSA)
 - Penicillinase-resistant penicillin (e.g. nafcillin, oxacillin) or cefazolin
 - Methicillin-resistant strains (MRSA)
 - Vancomycin or daptomycin + rifampin (activity against organisms in biofilms)

Mechanisms of antibiotic resistance(*Board relevant- pl review)

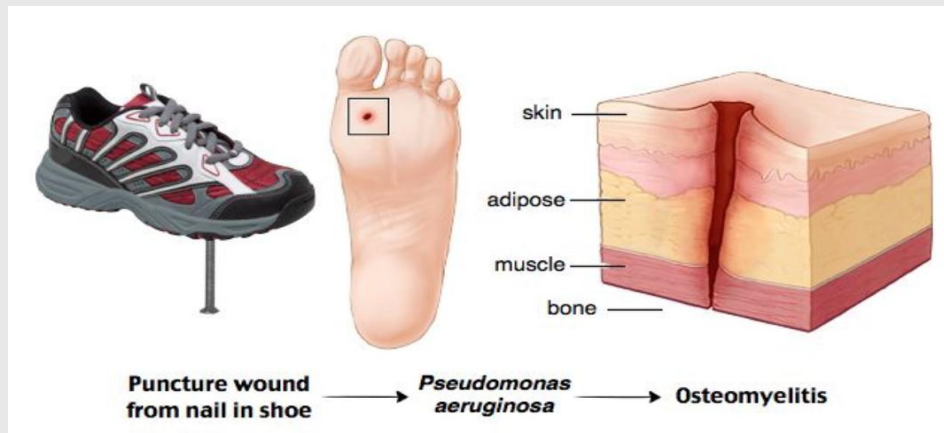
- ✓ Methicillin resistant *S. aureus* (MRSA)
 - Altered penicillin-binding proteins (PBPs) with low affinity to methicillin-mediated by ? Gene
- ✓ Vancomycin intermediate/resistant *S. aureus* (VISA/VRSA)
 - Altered cell wall(low affinity precursors-D-ala replaced by ?
- ✓ Penicillin resistance
 - Enzymatic inactivation(? Enzyme)
- ✓ Gentamicin resistance
 - Aminoglycoside-modifying enzymes and efflux

S. epidermidis and *Pseudomonas* spp.

- *Staphylococcus epidermidis* (biofilms)-bone/ joint surgeries, prosthetic joints)

- Gram-negative rods

★ *Pseudomonas* is an important cause in diabetic patients and IV drug abusers.



GNB
Oxidase positive
Fruity odor
Green pigments
Virulence factors-
Collagenases, Exotoxin A

Others GNB -*Escherichia coli*, *Serratia*, *Haemophilus influenzae*

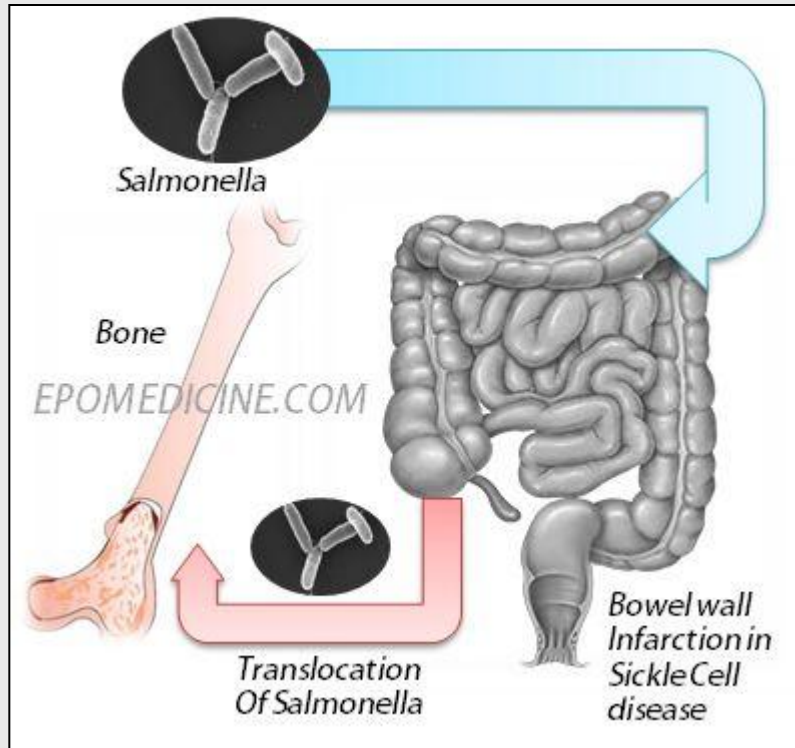
Treatment- Fluoroquinolones(high bone penetration)

Salmonella

- *Salmonella enterica* (typhoidal or non typhoidal types) are important causes of osteomyelitis esp. in patients with sickle cell anemia.

Gram negative bacilli
Member of Enterobacteriaceae
Oxidase negative
Non-Lactose Fermenters (NLF)
Produce H₂S
Motile

Treatment-Third generation cephalosporin

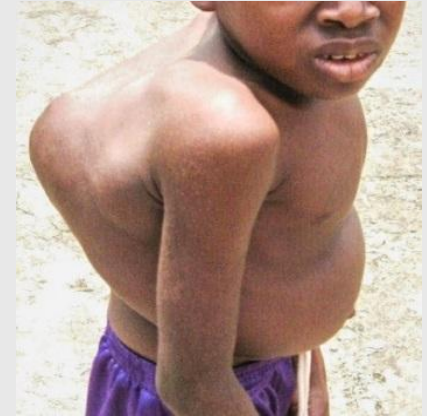


(More details of Salmonella will be taught in IS-GERD-GIT infections)

OTHER BACTERIA

- *Mycobacterium tuberculosis*

- **Pott's disease** (vertebrae and surrounding tissue)



- *Treponema pallidum*

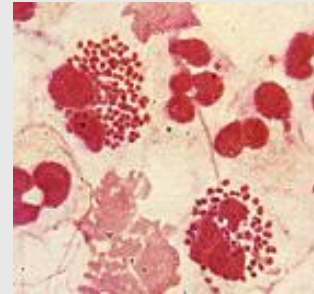
- Spirochete transmitted sexually or congenitally.
- Skeletal **syphilis**
- **Saber shin** of congenital syphilis.



Note-(Microbe details will be taught in
IS-GERD-Repro module-STI)

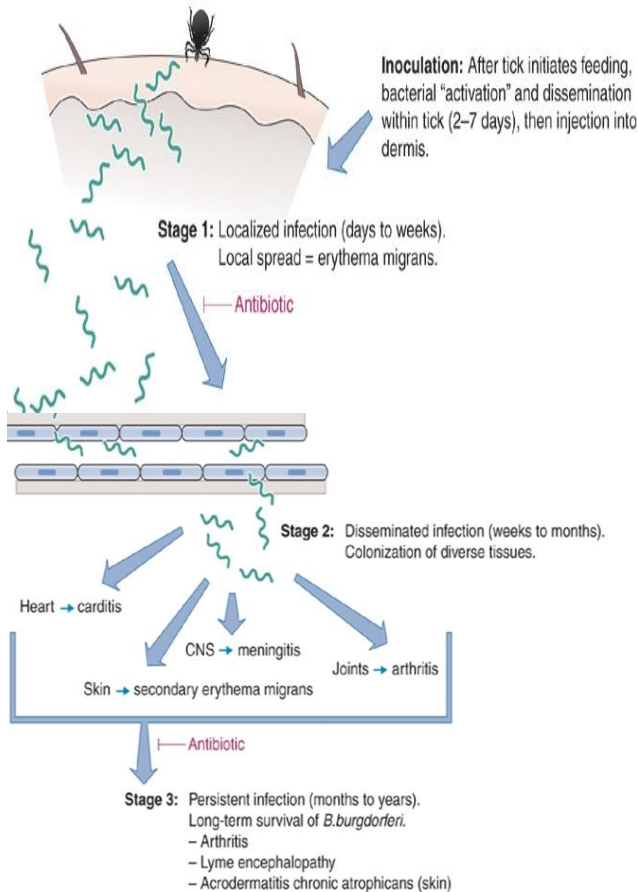
INFECTIOUS ARTHRITIS PATHOGENS

- *Neisseria gonorrhoeae* (Will be taught in IS-GERD-Repro-STI)
 - **Facultative intracellular kidney-bean shaped GNDC**
 - Disseminated gonococcal infection
 - More common in women — untreated asymptomatic infection
 - **Leading cause of purulent arthritis in sexually active age group**
 - Wrist, knees, ankles
- *Borrelia burgdorferi* (Recap from IS-FOM)
 - **Spirochete** transmitted by **Ixodes tick**
 - Virulence = **Major outer surface proteins and LPS**
 - Dissemination to joints → fluctuating arthritis, usually in knee
 - Usually seen in late stages
 - Intense inflammation due to immune response and auto immune response



Recap- Systemic Infections taught in FOM course

LYME DISEASE - PATHOGENESIS



Virulence factors of *Borrelia burgdorferi*- **Major Outer Surface Proteins and Endotoxins** play a role in pathogenesis by inducing inflammation and autoimmunity

Localised infection
Flu-like symptoms, arthritis,
Bull's eye rash

Disseminated infection
Carditis, meningitis, Bell's palsy, arthritis

Persistent Infection
Arthritis, encephalopathy etc.

LYME DISEASE – DIAGNOSIS

- Clinical findings + history of possible tick bite
- Serological tests
 - Recommended for later stages of disease
 - May be falsely negative in early disease
 - IgM antibodies detectable 2 to 4 weeks after initial infection (early)
 - IgG antibodies detectable 4 to 6 weeks after initial infection (late)
- **Western Blot**
 - Can detect IgM (early) and/or IgG (late)
- Microscopy:
 - Gram Stain**
 - **Large spirochetes**
 - Weakly staining GN
 - **Not recommended** for identification
 - Dark-field microscopy**
of *B. burgdorferi*

Treatment: Doxycycline or amoxicillin

Prevention- Prevention of tick bites: protective clothing, insect repellents...

Careful inspection and removal of crawling or attached ticks

REACTIVE ARTHRITIS

- Swelling and pain triggered by an infection in another part of the body
- Mostly follows an infection of the intestines or genitals or urethra.
- Usually targets knee, ankle and foot joints
- More likely in people with HLA-B27 gene
- May be associated with inflammation of eyes, skin and urethra (earlier k/a Reiter's Syndrome)
- Important bacteria causing Reactive Arthritis:
 - *Chlamydia trachomatis* (causes sexually transmitted infection)
 - *Shigella dysenteriae*
 - *Campylobacter jejuni*
 - *Salmonella spp.*
 - *Yersinia enterocolitica*

} cause Diarrhea and post infectious reactive arthritis
- *Borrelia burgdorferi*-Lyme disease can also be associated with Reactive arthritis
- Gp A streptococci-Rheumatic arthritis is an immune complex mediated arthritis
- Several viruses also cause arthritis- Rubella virus, Parvovirus, Dengue virus, Chikungunya V, Hepatitis B and C viruses (in some it is due to direct invasion while in others it is mediated through immune complex formation).

BUG LIST FOR MSK INFECTIONS- A & B (FOR MICROBIOLOGY EXAM)

Bacteria	Parasites	Fungi
<i>Staphylococcus aureus and epidermidis</i>	<i>Trichinella spiralis</i>	<i>Candida</i>
<i>Pseudomonas</i>	<i>Taenia solium-Cysticercosis</i>	
<i>Salmonella</i> *		
<i>Neisseria gonorrhoeae</i> *		
<i>Borrelia burgdorferi</i>		
<i>Streptococcus pyogenes</i> (GAS)		
<i>Clostridium perfringens</i>		
<i>Clostridium tetani</i>		

* Microbial details for *Salmonella* and *N.gonorrhoeae*(culture media, identification tests, virulence factors etc.) will not be tested. Only the morphological appearance and the epidemiological setting for these organisms may be tested.

THANK YOU

